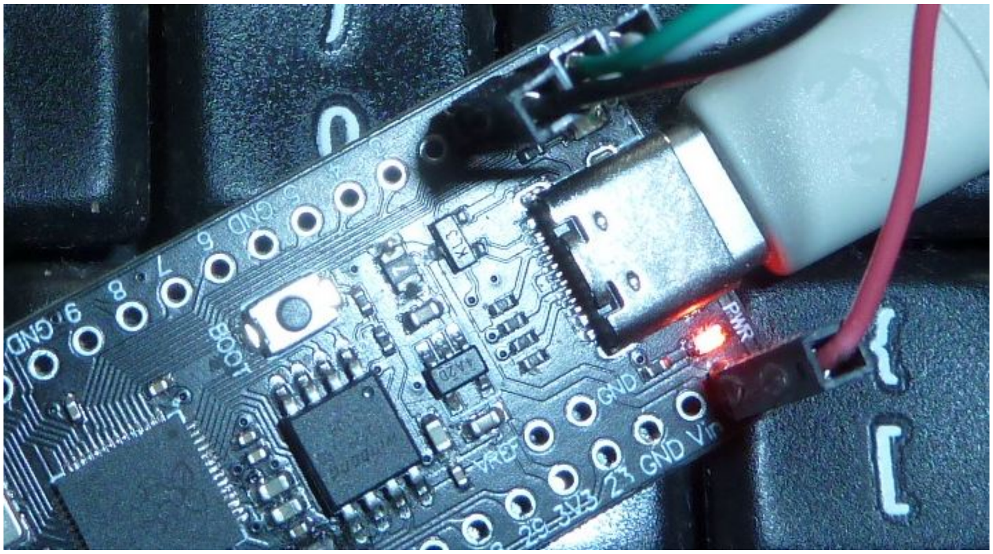
september 2026

noForth website

Pico RP2350 dev boards
with noForth t(v)(#) (solo/duo)

In this text we refer to these two documents:
- RP2350 datasheet
- Armv8-M Architecture Reference Manual

1. RP2350 Pico dev boards

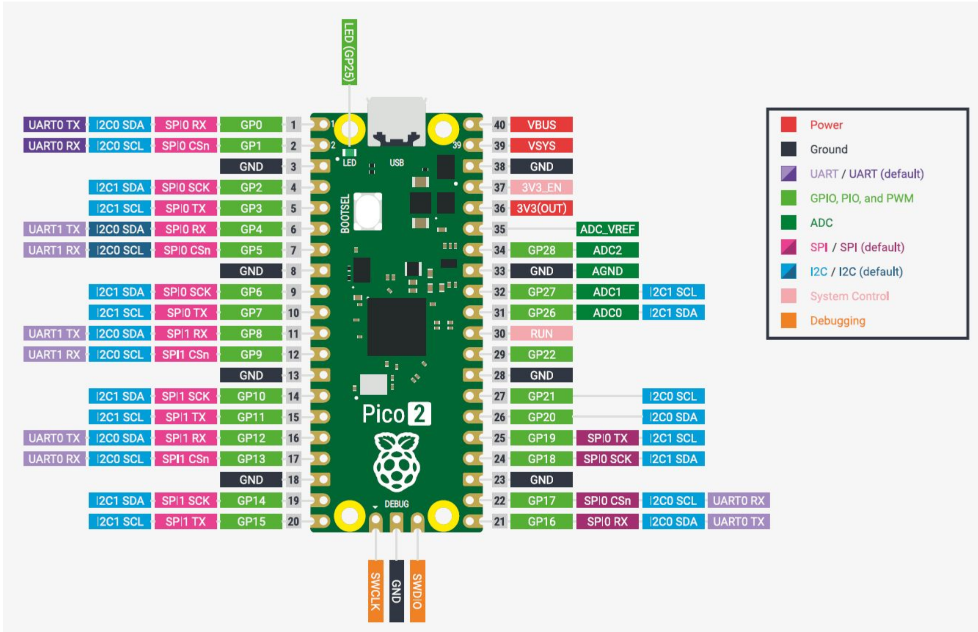


Core : M33 core with ARMv8-M ISA
Kit Contents: Various Pico Dev Boards

- Antratek: Pico 2 dev board
- Kiwi: Pico 2 dev board
- Olimex RP2350-PICO2-XXL board
- Pimoroni Pico Plus 2
- Waveshare RP2350-zero
- Waveshare RP2350-Plus
- Waveshare RP2350-PiPico

i/o port connections pico

```
GPIO
GPIO0  - UART0 TX        GPIO16 - ...
GPIO1  - UART0 RX        GPIO17 - ...
GPIO2  - ...             GPIO18 - ...
GPIO3  - ...             GPIO29 - ...
GPIO4  - UART1 TX        GPIO20 - ...
GPIO5  - UART1 RX        GPIO21 - ...
GPIO6  - ...             GPIO22 - ...
GPIO7  - ...             GPIO23 - SMPS
GPIO8  - ...             GPIO24 - VBUS sense
GPIO9  - ...             GPIO25 - LED
GPIO10 - ...             GPIO26 - ...
GPIO11 - ...             GPIO27 - ...
GPIO12 - ...             GPIO28 - ...
GPIO13 - ...             GPIO29 - VSYS
GPIO14 - ...
GPIO15 - ...
```



Pico2 board

Connectors

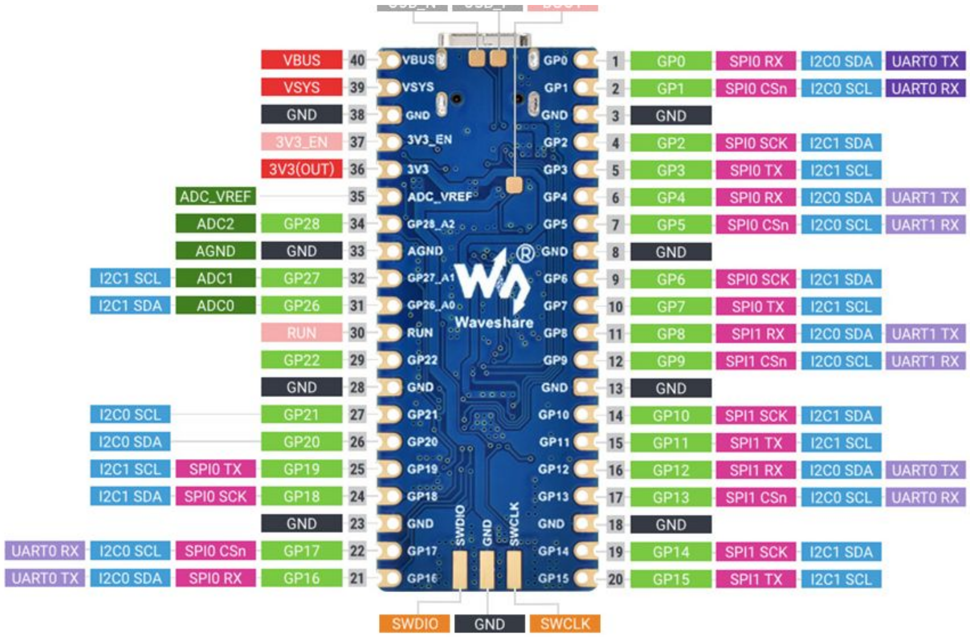
```
J0      = i/o GPIO and GND
        = i/o GPIO, RUN, GND, 5V, 3V3, 3V3_EN and ADC_VREF
J1      = USB-C Programming connector
J2      = Debug (SWD) port
```

Hardware Pico

- LED on GPIO25
- Switch BOOTSEL on QSPI_SS
- SPI Flash (16 to 128 Mbit)
- 12MHz crystal

Hardware RP2350-Plus

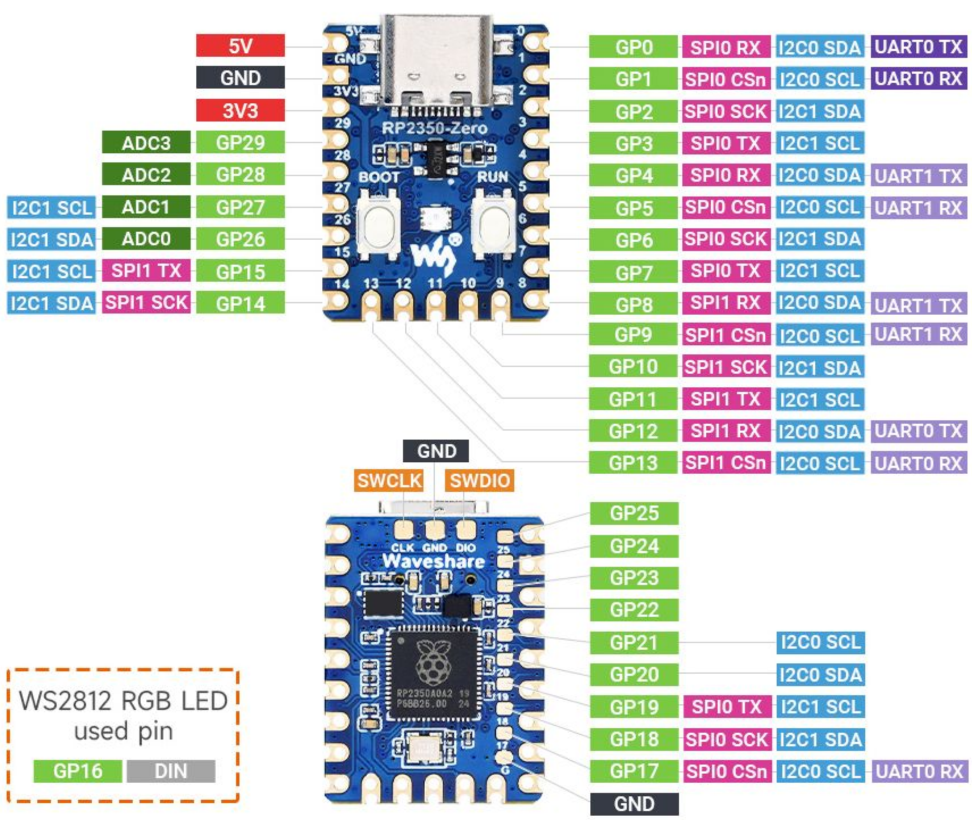
- LED on GPIO25, Lithium battery connector
- Two switches BOOTSEL on QSPI_SS & reset
- SPI Flash (128 Mbit=16 MByte)
- 12MHz crystal



RP2350-Plus board

Hardware RP2350-zero

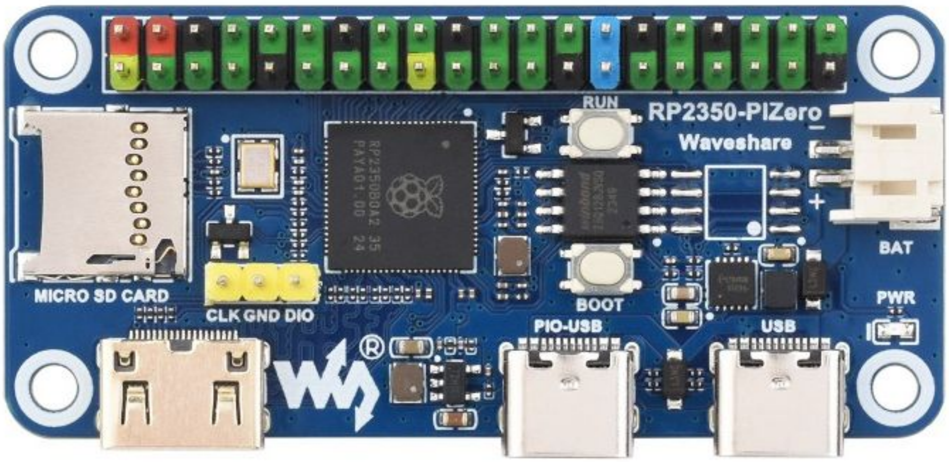
- WS2812 LED on GPIO16
- Two switches BOOTSEL on QSPI_SS & reset
- SPI Flash (16 Mbit)
- 12MHz crystal



RP2350-zero board

Hardware RP2350 PiZero

- Lithium battery connector
- Micro SD card connector
- Onboard DVI interface
- PIO-USB port
- Two switches BOOTSEL on QSPI_SS & reset
- SPI Flash (128 Mbit)
- 12MHz crystal



RP2350 PiZero board

2. I/O ports

I/O port addresses

The port registers are memory mapped.
First the base addresses for the I/O-port:

```
SIO base    = D0000000
User bank   = 40028000 (Two cells for each pin)
```

Port register offsets for SIO base

More functionality can be found from page 42ff.

```
Label           SIO base      Function
CPUID              00      Processor core ID
GPIO_IN            04      Input value for GPIO
GPIO_HI_IN         08      Input value for GPIO & QSPI
GPIO_OUT           10      Output value
GPIO_HI_OUT        14      Output value high
GPIO_OUT_SET       18      Output value set
GPIO_HI_OUT_SET    1C      Output value set high
GPIO_OUT_CLR       20      Output value clear
GPIO_HI_OUT_CLR    24      Output value clear high
GPIO_OUT_XOR       28      Output value toggle
GPIO_HI_OUT_XOR    2C      Output value toggle high
Etc.
```

Values for port control IO_BANK0_BASE registers

On page 590 to 593 of the RP2350 datasheet, you find a complete list of all functions that can be chosen for each GPIO pin. The pin control functions starts at offset+4 for GPIO0 and are increased by 8 for each next pin.

	Function								
GPIO	F1	F2	F3	F4	F5	F6	F7	F8	F9
0	SPI0 RX	UART0 TX	I2C0 SDA	PWM0 A	SIO	PIO0	PIO1		USB OVCUR DET
1	SPI0 CSn	UART0 RX	I2C0 SCL	PWM0 B	SIO	PIO0	PIO1		USB VBUS DET
2	SPI0 SCK	UART0 CTS	I2C1 SDA	PWM1 A	SIO	PIO0	PIO1		USB VBUS EN
3	SPI0 TX	UART0 RTS	I2C1 SCL	PWM1 B	SIO	PIO0	PIO1		USB OVCUR DET
4	SPI0 RX	UART1 TX	I2C0 SDA	PWM2 A	SIO	PIO0	PIO1		USB VBUS DET

```
1:  SPI-0 or 1
2:  UART-0 or 1
3:  I2C-0 or 1
4:  PWM-0 to PWM-7
5:  Standard I/O (Reset state)
6:  Programmable I/O (PIO-0)
7:  Programmable I/O (PIO-1)
8:  Clock in or out
9:  USB-bus
```

The reset value for all port control registers is 1F. More info in the 'RP2350 datasheet.PDF' from page 595ff.

3. RS232/USB driver

The USB chip for the SEED board is a dongle with the PL2303TA Prolific USB-chip. This chip needs a specific driver under Windows XP/7/8/10/11. Execute this file: PL2303-Prolific_DriverInstaller_v1200.exe. The default baudrate for the RP2350 controller with noForth is 115200 baud.

There is also a USB CDC driver available that runs on top of noForth tv# . This is a retrofittable option. An off-the-shelf UF2 is also available for simplicity.

4a. noForth t solo (multitask) memory map

RAM 20000000 - 21080000

```
20000000 IVECS      \ 48 vectors
     ... IVECS/     \ Vector table end
200000D0 HOT        \ Uhere starts here
20000200 ORIGIN     \ HERE starts here
\ ... system ...
2000xxxx HERE       \ Current systems end
2003F800 ...        \ Systems end
2003F800 FLYBUF     \ Flyer buffer (400 bytes)
2003FC00 FLYBUF/    \ End of flyer buffer
2003FD00 SP0        \ Data stack (100 bytes grows down)
2003FF00 RP0        \ Return stack (200 bytes grows down)
2003FF00 TIB        \ Input buffer (100 bytes)
20040000 TIB/
20080000 MEMTOP     \ End of RAM memory
```

RAM 20080000 - 20082000

8 kByte of free RAM memory in two 4 kByte banks.
The multitasker version uses a part of this memory for new tasks.

4b. noForth t duo (multitask) memory map

noForth t memory map core-0
RAM 20000000 - 20040000

```
20000000 IVECS      \ 48 vectors
     ... IVECS/     \ Vector table end
200000D0 HOT        \ Uhere starts here
20000200 ORIGIN     \ HERE starts here
\ ... system ...
2000xxxx HERE       \ Current systems end
2003F800 ...        \ Systems end
2003F800 FLYBUF     \ Flyer buffer (400 bytes)
2003FC00 FLYBUF/    \ End of flyer buffer
2003FD00 SP0        \ Data stack (100 bytes grows down)
2003FF00 RP0        \ Return stack (200 bytes grows down)
2003FF00 TIB        \ Input buffer (100 bytes)
20040000 TIB/
20040000 MEMTOP     \ End of core-0 RAM memory
```

noForth t memory map core-1

RAM 20040000 – 20080000

```
20040000 IVECS      \ 48 vectors
     ... IVECS/     \ Vector table end
200400D0 HOT        \ Uhere starts here
20040200 ORIGIN     \ HERE starts here
\ ... system ...
2004xxxx HERE       \ Current systems end
2007F800 ...        \ Systems end
2007F800 FLYBUF     \ Flyer buffer (400 bytes)
2007FC00 FLYBUF/    \ End of flyer buffer
2007FD00 SP0        \ Data stack (100 bytes grows down)
2007FF00 RP0        \ Return stack (200 bytes grows down)
2007FF00 TIB        \ Input buffer (100 bytes)
20080000 TIB/
20080000 MEMTOP     \ End of core-1 RAM memory
```

RAM 20080000 – 20082000

8 kByte of free RAM memory in two 4 kByte banks.
The multitasker version uses a part of this memory for new tasks.

4c. FLASH ROM (XIP mode) 10000000 – 103FFFFF

```
 10000000  Secundairy boot code, loads image-1 on startup
 10000100  Image-1  \ Cold start image invoked by COLD to
 10081000  Image-2  \ Spare image invoked by COLD2
 10101000  Free     \ Start of free flash space
```

- TIB/ is a changeable uvalue (Udata).
- Udata is saved by FREEZE too.
- At start-up (and reset or COLD) noForth moves from Flash to RAM.

5. Interrupt vector table

See 'RP2350 datasheet.PDF' for more details (page 82ff) and 'ARMv6-M Architecture Reference Manual.pdf' from page 218ff. The table is pointed to by the VTOR register at PPB_BASE + ED08. Note that the low 8 bits of VTOR are always zero , so it's 256 bytes page aligned.

RP2350 exceptions:

```
0000  00 - Length of noForth image
0004  01 - Reset vector
0008  02 - NMI handler
000C  03 - Hard fault handler
0010  04 - MPU fault handler
0014  05 - Reserved
0018  06 - Reserved
001C  07 - Reserved
0020  08 - Reserved
0024  09 - Reserved
0028  10 - Reserved
002C  11 - Supervisor call handler
0030  12 - Reserved
0034  13 - Reserved
0038  14 - Pend supervidor handler
003C  15 - Systick handler
```

RP2350 NVIC IRQ interrupts:

```
0040  00 - Timer0 IRQ 0         00AC  27 - SIO IRQ FIFO NS
0044  01 - Timer0 IRQ 1         00B0  28 - SIO IRQ BELL NS
0048  02 - Timer0 IRQ 2         00B4  29 - SIO IRQ MTIMECMP
004C  03 - Timer0 IRQ 3         00B8  30 - Clock IRQ
0050  04 - Timer1 IRQ 0         00BC  31 - SPI 0 IRQ
0054  05 - Timer1 IRQ 1         00C0  32 - SPI 1 IRQ
0058  06 - Timer1 IRQ 2         00C4  33 - UART 0 IRQ
005C  07 - Timer1 IRQ 3         00C8  34 - UART 1 IRQ
0060  08 - PWM IRQ wrap 0       00CC  35 - ADC IRQ fifo
0064  09 - PWM IRQ wrap 1       00D0  36 - I2C 0 IRQ
0068  10 - DMA IRQ 0            00D4  37 - I2C 1 IRQ
006C  11 - DMA IRQ 1            00D8  38 - OTP IRQ
0070  12 - DMA IRQ 2            00DC  39 - TRNG IRQ
0074  13 - DMA IRQ 3            00E0  40 - PROC0_IRQ_CTI
0078  14 - USB control IRQ      00E4  41 - PROC1_IRQ_CTI
007C  15 - PIO 0 IRQ 0          00E8  42 - PLL_SYS_IRQ
0080  16 - PIO 0 IRQ 1          00EC  43 - PLL_USB_IRQ
0084  17 - PIO 1 IRQ 0          00F0  44 - POWMAN_IRQ_POW
0088  18 - PIO 1 IRQ 1          00F4  45 - POWMAN_IRQ_TIMER
008C  19 - PIO 2 IRQ 0          00F8  46 - SPAREIRQ_IRQ_0
0090  20 - PIO 2 IRQ 1          00FC  47 - SPAREIRQ_IRQ_1
0094  21 - IO IRQ bank 0        0100  48 - SPAREIRQ_IRQ_2
0098  22 - IO IRQ bank 0 NS     0104  49 - SPAREIRQ_IRQ_3
009C  23 - IO IRQ QSPI          0108  50 - SPAREIRQ_IRQ_4
00A0  24 - IO IRQ QSPI NS       010C  51 - SPAREIRQ_IRQ_5
00A4  25 - SIO IRQ FIFO
00A8  26 - SIO IRQ BELL
```